

1 Single Neuron/Perceptron

$$\sigma \left(\sum_{i=1}^n X_i W_i + b \right) \quad (1)$$

2 Multi layer Perceptron

2.1 Notations

W_{jk}^l is weight from k index in row $l-1 \rightarrow j$ index in l in layer l
Lets say we have the below matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

then it is a matrix with j in current layer rows and k in previous layer columns
and W_{jk} will be a node to node weight.

b_j^l is bias for layer l at index j in layer l .

2.2 Feed-forward

$$c = (a^1 - a^0)^2 \quad (2)$$

2.3 Feed-forward

$$a^1 = \sigma(w^1 a^0 + b^1) \quad (3)$$

$$z^1 = w^1 a^0 + b^1 \quad (4)$$

$$a^1 = \sigma(z) \quad (5)$$

2.4 Back propogation

$$\frac{d(\sigma(x))}{dx} = (\sigma(x))(1 - \sigma(x)) \quad (6)$$

$$\frac{\partial c}{\partial w^1} = \frac{\partial c}{\partial a^1} \frac{\partial a^1}{\partial z^1} \frac{\partial z^1}{\partial w^1} \quad (7)$$

$$\frac{\partial c}{\partial w^1} = 2 * (a^1 - a^0) * (a^1) * (1 - a^1) * a^0 \quad (8)$$

$$\frac{\partial c}{\partial b^1} = \frac{\partial c}{\partial a^1} \frac{\partial a^1}{\partial z^1} \frac{\partial z^1}{\partial b^1} \quad (9)$$

$$\frac{\partial c}{\partial b^1} = 2 * (a^1 - a^0) * (a^1) * (1 - a^1) * 1 \quad (10)$$

$$\frac{\partial c}{\partial a^0} = \frac{\partial c}{\partial a^1} \frac{\partial a^1}{\partial z^1} \frac{\partial z^1}{\partial a^0} \quad (11)$$

$$\frac{\partial c}{\partial a^0} = 2 * (a^1 - a^0) * (a^1) * (1 - a^1) * w^1 \quad (12)$$